

UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS

GCE Advanced Subsidiary Level and GCE Advanced Level

**MARK SCHEME for the October/November 2011 question paper
for the guidance of teachers**

9702 PHYSICS

9702/41

Paper 4 (A2 Structured Questions), maximum raw mark 100

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes must be read in conjunction with the question papers and the report on the examination.

- Cambridge will not enter into discussions or correspondence in connection with these mark schemes.

Cambridge is publishing the mark schemes for the October/November 2011 question papers for most IGCSE, GCE Advanced Level and Advanced Subsidiary Level syllabuses and some Ordinary Level syllabuses.



Page 2	Mark Scheme: Teachers' version	Syllabus	Paper
	GCE AS/A LEVEL – October/November 2011	9702	41

Section A

- 1 (a) gravitational force provides the centripetal force
 $GMm/r^2 = mr\omega^2$ (must be in terms of ω)
 $r^3\omega^2 = GM$ and GM is a constant B1 B1 B1 [3]
- (b) (i) 1. for Phobos, $\omega = 2\pi/(7.65 \times 3600)$
 $= 2.28 \times 10^{-4} \text{ rad s}^{-1}$
 $(9.39 \times 10^6)^3 \times (2.28 \times 10^{-4})^2 = 6.67 \times 10^{-11} \times M$
 $M = 6.46 \times 10^{23} \text{ kg}$ C1 A1 [3]
2. $(9.39 \times 10^6)^3 \times (2.28 \times 10^{-4})^2 = (1.99 \times 10^7)^3 \times \omega^2$
 $\omega = 7.30 \times 10^{-5} \text{ rad s}^{-1}$
 $T = 2\pi/\omega = 2\pi/(7.30 \times 10^{-5})$
 $= 8.6 \times 10^4 \text{ s}$
 $= 23.6 \text{ hours}$ C1 C1 A1 [3]
- (ii) *either* almost 'geostationary'
or satellite would take a long time to cross the sky B1 [1]
- 2 (a) e.g. moving in random (rapid) motion of molecules/atoms/particles
no intermolecular forces of attraction/repulsion
volume of molecules/atoms/particles negligible compared to volume of container
time of collision negligible to time between collisions
(1 each, max 2) B2 [2]
- (b) (i) 1. number of (gas) molecules B1 [1]
2. mean square speed/velocity (of gas molecules) B1 [1]
- (ii) *either* $pV = NkT$ *or* $pV = nRT$ and links n and k
and $\langle E_K \rangle = \frac{1}{2}m\langle c^2 \rangle$ M1
- clear algebra leading to $\langle E_K \rangle = \frac{3}{2}kT$ A1 [2]
- (c) (i) sum of potential energy and kinetic energy of molecules/atoms/particles
reference to random (distribution) M1 A1 [2]
- (ii) no intermolecular forces so no potential energy B1
(change in) internal energy is (change in) kinetic energy and this is
proportional to (change in) T B1 [2]

Page 3	Mark Scheme: Teachers' version	Syllabus	Paper
	GCE AS/A LEVEL – October/November 2011	9702	41

- 3 (a) (i) amplitude remains constant B1 [1]
- (ii) amplitude decreases gradually
light damping M1
A1 [2]
- (iii) period = 0.80 s C1
frequency = 1.25 Hz (*period not 0.8 s, then 0/2*) A1 [2]
- (b) (i) (induced) e.m.f. is proportional to
rate of change/cutting of (magnetic) flux (linkage) M1
A1 [2]
- (ii) a current is induced in the coil M1
as magnet moves in coil A1
current in resistor gives rise to a heating effect M1
thermal energy is derived from energy of oscillation of the magnet A1 [4]
- 4 (a) (i) zero field (strength) inside spheres B1 [1]
- (ii) *either* field strength is zero
or the fields are in opposite directions
at a point between the spheres M1
A1 [2]
- (b) (i) field strength is (–) potential gradient (*not* V/x) B1 [1]
- (ii) 1. field strength has maximum value
at $x = 11.4$ cm B1
B1 [2]
2. field strength is zero B1
either at $x = 7.9$ cm (*allow* ± 0.3 cm)
or at 0 to 1.4 cm *or* 11.4 cm to 12 cm B1 [2]
- 5 (a) (i) $Bqv(\sin\theta)$ or $Bqv(\cos\theta)$ B1 [1]
- (ii) qE B1 [1]
- (b) F_B must be opposite in direction to F_E
so magnetic field into plane of paper B1
B1 [2]

Page 4	Mark Scheme: Teachers' version	Syllabus	Paper
	GCE AS/A LEVEL – October/November 2011	9702	41

- 6 (a) (i) period = $1/50$
 $t_1 = 0.03 \text{ s}$ C1
A1 [2]
- (ii) peak voltage = 17.0 V A1 [1]
- (iii) r.m.s. voltage = $17.0/\sqrt{2}$
= 12.0 V A1 [1]
- (iv) mean voltage = 0 A1 [1]
- (b) power = V^2/R C1
= $12^2/2.4$
= 60 W A1 [2]
- 7 (a) each line represents photon of specific energy M1
photon emitted as a result of energy change of electron M1
specific energy changes so discrete levels A1 [3]
- (b) (i) arrow from -0.85 eV level to -1.5 eV level B1 [1]
- (ii) $\Delta E = hc/\lambda$ C1
= $(1.5 - 0.85) \times 1.6 \times 10^{-19}$ C1
= $1.04 \times 10^{-19} \text{ J}$
 $\lambda = (6.63 \times 10^{-34} \times 3.0 \times 10^8)/(1.04 \times 10^{-19})$
= $1.9 \times 10^{-6} \text{ m}$ A1 [3]
- (c) spectrum appears as continuous spectrum crossed by dark lines B1
two dark lines B1
electrons in gas absorb photons with energies equal to the excitation energies M1
light photons re-emitted in all directions A1 [4]
- 8 (a) (i) time for initial number of nuclei/activity M1
to reduce to one half of its initial value A1 [2]
- (ii) $\lambda = \ln 2/(24.8 \times 24 \times 3600)$ M1
= $3.23 \times 10^{-7} \text{ s}^{-1}$ A0 [1]
- (b) (i) $A = \lambda N$ C1
 $3.76 \times 10^6 = 3.23 \times 10^{-7} \times N$
 $N = 1.15 \times 10^{13}$ A1 [2]
- (ii) $N = N_0 e^{-\lambda t}$ C1
= $1.15 \times 10^{13} \times \exp(-\{\ln 2 \times 30\}/24.8)$ A1 [2]
= 4.97×10^{12}
- (c) ratio = $(4.97 \times 10^{12})/(1.15 \times 10^{13} - 4.97 \times 10^{12})$ C1
= 0.76 A1 [2]

Page 5	Mark Scheme: Teachers' version	Syllabus	Paper
	GCE AS/A LEVEL – October/November 2011	9702	41

Section B

- 9 (a) e.g. reduced gain
increased stability
greater bandwidth or less distortion
(allow any two sensible suggestions, 1 each, max 2) B2 [2]
- (b) (i) V^- connected to midpoint between resistors
 V_{OUT} clear and input to V^+ clear B1
B1 [2]
- (ii) gain = $1 + R_F/R$
 $15 = 1 + 12000/R$ C1
 $R = 860 \Omega$ A1 [2]
- (c) graph: straight line from (0,0) to (0.6,9.0) B1
straight line from (0.6,9.0) to (1.0,9.0) B1 [2]
- (d) either relay can be used to switch a large current/voltage
output current of op-amp is a few mA/very small M1
or relay can be used as a remote switch A1 [2]
for inhospitable region/avoids using long heavy cables (M1)
(A1)
- 10 (a) e.g. large bandwidth/carries more information
low attenuation of signal
low cost
smaller diameter, easier handling, easier storage, less weight
high security/no crosstalk
low noise/no EM interference
(allow any four sensible suggestions, 1 each, max 4) B4 [4]
- (b) (i) infra-red B1 [1]
- (ii) lower attenuation than for visible light B1 [1]
- (c) (i) gain/dB = $10 \lg(P_2/P_1)$ C1
 $26 = 10 \lg(P_2/9.3 \times 10^{-6})$
 $P_2 = 3.7 \times 10^{-3} \text{ W}$ A1 [2]
- (ii) power loss along fibre = $30 \times 0.2 = 6.0 \text{ dB}$ C1
either $6 = 10 \lg(P/3.7 \times 10^{-3})$ or $6 \text{ dB} = 4 \times 3.7 \times 10^{-3}$
or $32 = 10 \lg(P/9.3 \times 10^{-6})$
input power = $1.5 \times 10^{-2} \text{ W}$ A1 [2]

Page 6	Mark Scheme: Teachers' version	Syllabus	Paper
	GCE AS/A LEVEL – October/November 2011	9702	41

- 11 (a) (i) switch M1
so that one aerial can be used for transmission and reception A1 [2]
- (ii) tuning circuit M1
to select (one) carrier frequency (and reject others) A1 [2]
- (iii) analogue-to-digital converter/ADC M1
converts microphone output to a digital signal A1 [2]
- (iv) (a.f.) amplifier (*not r.f. amplifier*) M1
to increase (power of) signal to drive the loudspeaker A1 [2]
- (b) e.g. short aerial so easy to handle
short range so less interference between base stations
larger waveband so more carrier frequencies
(*any two sensible suggestions, 1 each, max 2*) B2 [2]